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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : IT-801D

CRYPTOGRAPHY & NETWORK SECURITY

Time Allotted : 3 Hours *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A
(Multiple Choice Type Questions)

1. Choose the correct alternatives for the following : $10 \times 1 = 10$

i) The principle of ..... ensures that only the sender and the intended recipient(s) have access to the contents of a message.

a) authentication

b) confidentiality

c) integrity

d) none of these.

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ii) RSA ..... be used for Digital Signatures.

a) must not

b) cannot

c) can

d) should not.

iii) While creating an envelope, we encrypt the ..... with the .....

a) sender's private key, one time session key

b) receiver's public key, one time session key

c) one time session key, sender's private key

d) one time session key, receiver's public key.

iv) DOS attacks are caused by

a) alternation

b) authentication

c) fabrication

d) replay attacks.

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v) DES encrypts blocks of ..... bits.

a) 32

b) 56

c) 64

d) 128.

vi) Caesar Cipher is an example of

a) Substitution Cipher

b) Transposition Cipher

c) Substitution as well as Transposition Cipher

d) None of these.

vii) Which one of the following algorithms is not used in asymmetric-key cryptography ?

a) RSA algorithm

b) Diffie-hellman algorithm

c) Electronic code book algorithm

d) None of these.

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viii) Cryptographic hash function takes an arbitrary block of data and returns

a) fixed size bit string

b) variable size bit string

c) both (a) and (b)

d) none of these.

ix) In asymmetric key cryptography, the private key is kept by

a) sender

b) receiver

c) sender and receiver

d) all the connected devices to the network.

x) Cryptanalysis is used

a) to find some insecurity in a cryptographic scheme

b) to increase the speed

c) to encrypt the data

d) none of these.

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GROUP – B
(Short Answer Type Questions)

Answer any *three* of the following $3 \times 5 = 15$

2. What are cryptanalysis and cryptography ? Define threat and attack. 3 + 2

3. a) What are symmetric cipher and asymmetric cipher ?

   b) What are the drawbacks of symmetric cipher ?

   3 + 2

4. What are the typical contents of Digital Certificate ? Discuss key exchange protocol. 3 + 2

5. Explain DOS attack. What is IP sniffing and IP spoofing. 3 + 2

6. Explain Diffie-Hellman key exchange algorithm. 5

GROUP – C
(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

7. a) What are the four main stages in AES operation ?

   b) Let the state matrix in any intermediate stage of AES is

ac	21	34	ec
02	76	ea	02
13	50	46	a4
92	76	ab	Ba

   What would be the state matrix after the Shift Row Operation ?

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   c) In an AES cryptosystem the Round Key for Round 6 is

   EA BC 73 23 45 67 32 87 E2 3D 9B 02 33 4E A2 F0

   Find the first 4 bytes of the Round key for round 7. The values of RC [j] in Hexa-decimal is given as follows :

J	1	2	3	4	5	6	7	8	9	10
RC [j]	01	02	04	08	10	20	40	80	1B	36

   4 + 3 + 8

8. a) Briefly describe the RSA algorithm and show how it provides security to message / data.

   b) How digital signatures can be generated ? What does it provide to a message ? 9 + 6

9. a) What is a firewall ?

   b) What are the different types of firewall ? Briefly explain working principle of each.

   c) What are the limitations of firewall ?

   d) What is a worm ? How does it differ from a virus ?

   2 + 6 + 3 + 4

10. a) Describe SHA-1 (message digest) algorithm.

    b) How SHA-1 is differing from MD5 ?

    c) It is known that, in symmetric-key cryptography, it is difficult to share the same secret between two communicating parties. How can public-key cryptography solve this problem ?

    d) Give an example of a message authentication using symmetric-key and asymmetric-key cryptography.

    7 + 2 + 4 + 2

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11. Write short notes on any *three* of the following : $3 \times 5 = 15$

a) Intruders

b) Malicious programs

c) Digital signatures

d) Key management

e) Firewall.

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